

The development of university foundations in China

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Research question 1: What are the characteristics of university foundations in China?

Foundations at top Chinese universities are still very young. In the 39-university dataset, they are about 12 years old on average. At the national level, fewer than 20% of universities have a foundation: as of 2017, only 525 out of more than 3,000 higher education institutions had registered one.

Staffing is also very lean. In this sample, each foundation employs about 13 full-time staff on average. Across China, nearly half report only 1–5 full-time staff, and 79 report zero.

Spending patterns show a similar story. In 2018, program expenses account for about 97% of total expenses, while reported operation expenses take up only around 2% on average.

Why this matters: Chinese university foundations are young and run with very small, low-cost teams. This limits their capacity today but also creates room for universities to grow philanthropy by hiring more staff, building better systems, and tracking results over time.

Descriptive statistics for non-percentage variables of university foundations in China

Variable type: numeric

skim_variable	n_missing	mean	sd	p0	p25	p50	p75	p100
Years	0	12	5	3	8	11	14	24
Staff_number	0	13	11	0	6	10	16	49
Ranking	0	20	11	1	10	20	30	39
Graduate number	0	17533	6487	4708	13348	15829	19863	32143
Undergrad number	0	21563	10372	7124	15118	17724	25327	54216
Investment return(m, RMB)	0	28	74	0	0	4	19	422
Total invesment(m, RMB)	0	572	1485	0	1	70	366	7593
Total contributions(m, RMB)	0	180	388	3	36	60	112	2136

Descriptive statistics for percentage variables of university foundations in China

Variable type: numeric

skim_variable	n_missing	mean	sd	p0	p25	p50	p75	p100
ROI	0	0.04	0.05	0.00	0.00	0.03	0.05	0.19
Porject expense	0	0.22	0.20	0.06	0.12	0.18	0.24	0.96
Operation expense	0	0.02	0.03	0.00	0.01	0.02	0.03	0.14

Research question 2: What factors predict the total amount of contributions received by university foundations in China?

For RQ2, I examined how five variables shown in the tables and plots relate to total contributions. Correlation analysis identifies several statistically significant relationships with total contributions, and these variables are then evaluated together in a multiple linear regression model.

In that model, full-time staff and graduate students are the strongest positive predictors of total contributions, foundation age and ranking have weaker positive effects, and undergraduate enrollment is not significant. Together, these five variables explain about 73% of the variance in total contributions across the 39 foundations.

Why this matters: The results point to a clear, testable lever—foundations with more full-time staff tend to raise more money.

Descriptive statistics for key contribution predictors

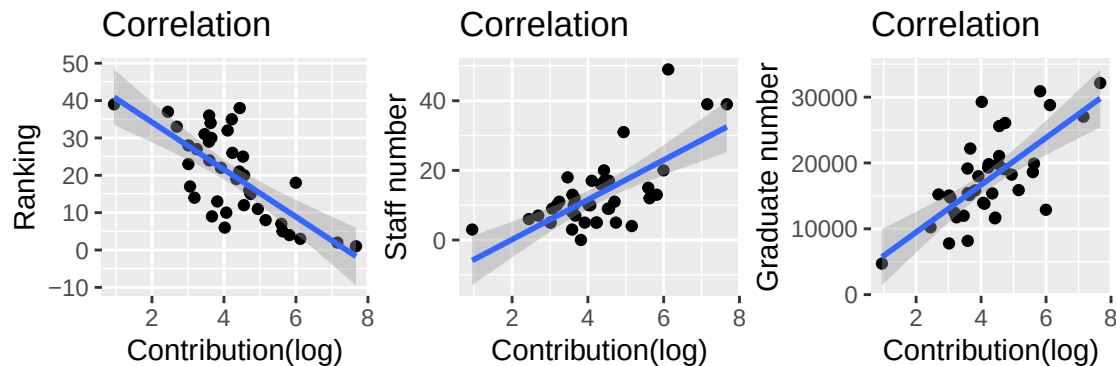
Variable type: numeric

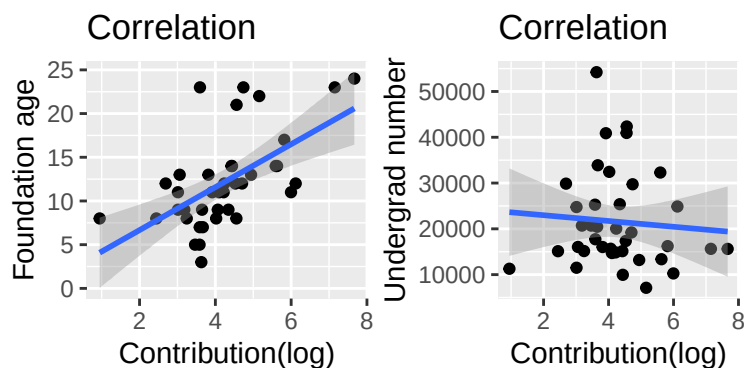
skim_variable	n_missing	mean	sd	p0	p25	p50	p75	p100
Years	0	12	5	3	8	11	14	24
Staff number	0	13	11	0	6	10	16	49
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Correlation matrix and p-values for key predictors

new_variables_1	new_variables_2	correlation	p_value
Graduate number	Ranking	-0.702139	0.000004
Staff number	Contribution	0.678052	0.000349
Graduate number	Contribution	0.547334	0.000012
Contribution	Years	0.541965	0.000027
Ranking	Contribution	-0.506918	0.000006

ggplot correlation plots for contributions and key predictors





Research question 3: How do Chinese university foundations differ from their U.S.counterparts?

The two case foundations invest in very different ways. Mulan University (MLU) follows a conservative strategy, puts most money into short-term products, and never shows a negative yearly ROI over the 10-year period; its average ROI is about 5% with only small year-to-year changes. The GPU foundation takes more risk, uses a more aggressive portfolio, and experiences both positive and negative years, with a higher average ROI of about 7% and much larger swings.

Staffing levels also differ sharply: MLU employs around 39 full-time staff on average, while the GPU foundation averages about 627.

Within China, most of the 39 top university foundations report low single-digit ROI and many show zero investment activity or zero returns, so MLU's average ROI near 5% is relatively high in this domestic group.

Why this matters: The comparison between MLU and GPU highlights a clear trade-off—accepting more investment risk with a much larger professional team can deliver higher, but more volatile, returns over time.

Descriptive statistics for non-percentage variables at Mulan University (10 years)

Variable type: numeric

skim_variable	n_missing	mean	sd	p0	p25	p50	p75	p100
Year range	0	2014	3	2010	2012	2014	2017	2019
Staff number	0	39	12	21	30	39	50	52
Investment return(m, RMB)	0	60	43	12	23	50	90	126
Total investments(m, RMB)	0	1325	890	126	627	1270	1903	2628
Total revenues(m, RMB)	0	486	272	219	318	405	569	1140
Contributions(m, RMB)	0	420	243	203	285	336	454	1013

Descriptive statistics for percentage variables at Mulan University (10 years)

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100
ROI	0	1	0.05	0.02	0.01	0.04	0.05	0.06	0.09

Descriptive statistics for non-percentage variables at GPU University (10 years)

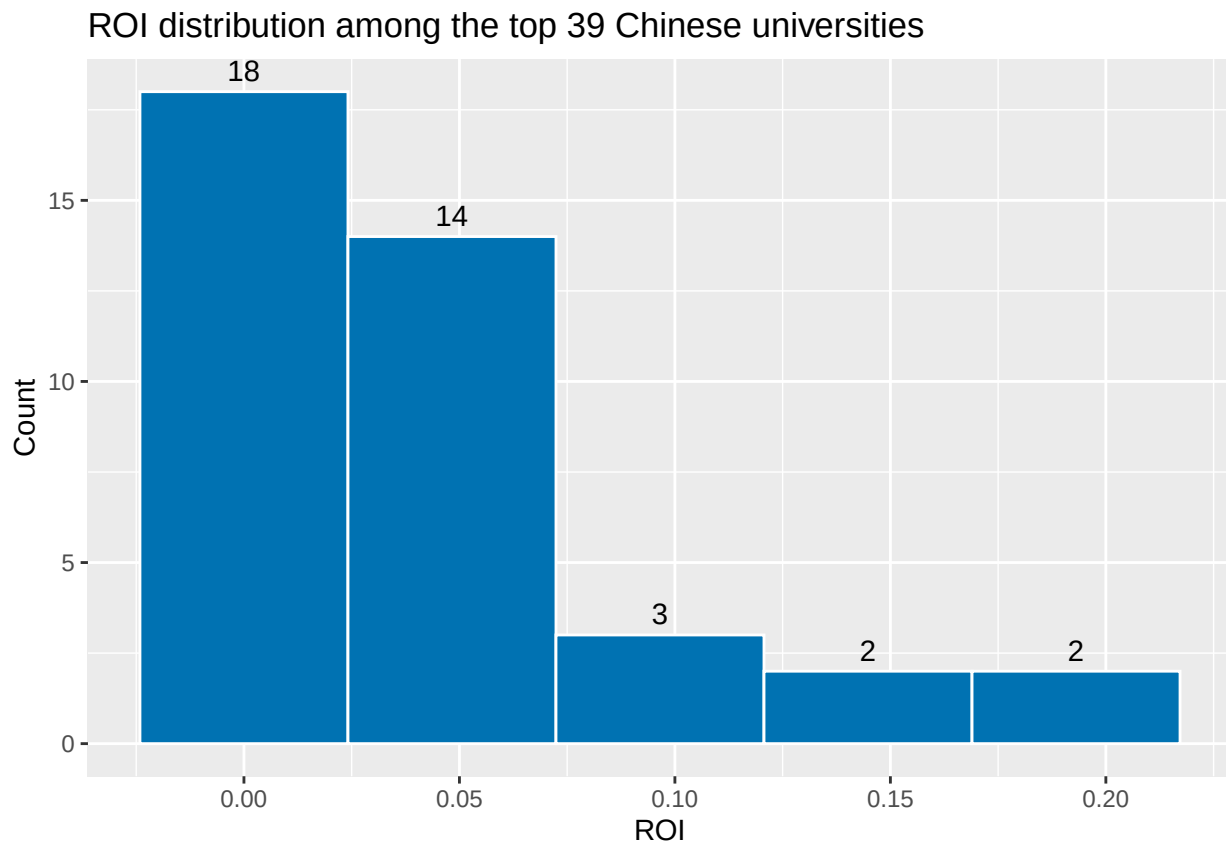
Variable type: numeric

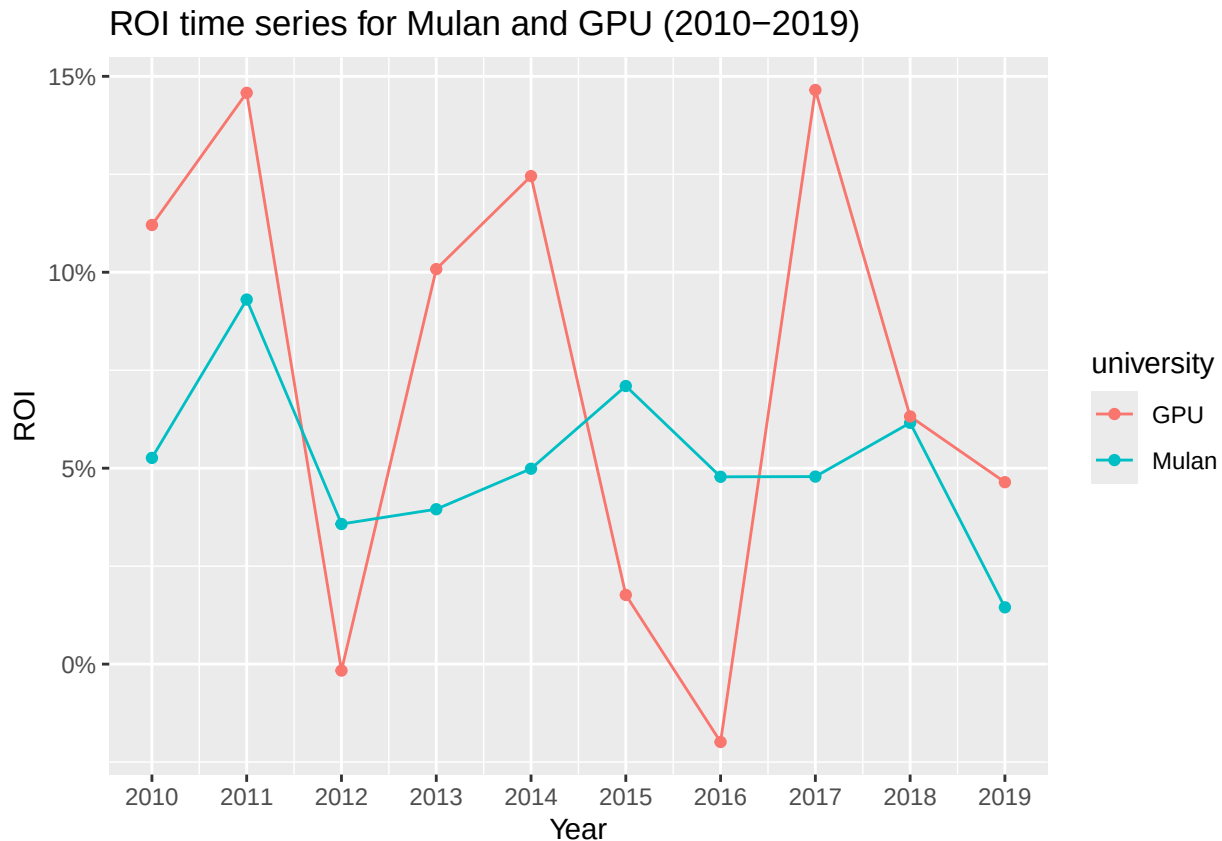
skim_variable	n_missing	mean	sd	p0	p25	p50	p75	p100
Year	0	2014	3	2010	2012	2014	2017	2019
Staff number	0	627	166	439	512	580	687	918
Contributions(m, USD)	0	104	46	58	66	87	137	182
Total revenues(m, USD)	0	168	81	64	129	152	198	344
Investment return(m, USD)	0	59	51	-16	24	64	85	156
Total investments(m, USD)	0	813	223	520	640	776	1001	1165

Descriptive statistics for percentage variables at GPU University (10 years)

Variable type: numeric

skim_variable	n_missing	mean	sd	p0	p25	p50	p75	p100
ROI	0	0.07	0.06	-0.02	0.02	0.08	0.12	0.15





Appendix

```
# Setup and package loading
appendix<-TRUE
knitr::opts_chunk$set(echo = FALSE, warning=FALSE, message=FALSE, appendix = TRUE)
library(tidyverse)
library(readxl)
library(skimr)
library(scales)
library(kableExtra)

# Import data from Excel
data <- read_excel("datasets/40_uni_data_only.xlsx",
  sheet = "40 Uni", skip = 1)
GPU <- read_excel("datasets/40_uni_data_only.xlsx",
  sheet = "ISU_R")
Mulan <- read_excel("datasets/40_uni_data_only.xlsx",
  sheet = "Mulan_R")

# Transform financial variables (RMB/USD to millions)
data <- data |>
  mutate(across(c(T_contributions, T_revenues, T_assets_last,
    Investment_return, T_investment), ~./1000000))
GPU <- GPU |>
  mutate(across(c(`Investment return`, Contributions,
    `Total investments`, `Total revenues`), ~./1000000))
```

```

Mulan <- Mulan |>
  mutate(across(c(`Investment return`, Contributions,
                  `Total revenues`, `Total investments`), ~./1000000))

# Explore dataset structure
data |>
  head()
data |>
  tail()
data |>
  summary()
data |>
  skim() |>
  select(-numeric.hist)

# Descriptive statistics for non-percentage variables
options(scipen = 999)
data |>
  select(years, No_ft_staff, ranking,
         No_Graduate_SS, No_Undergra,
         Investment_return, T_investment,
         T_contributions, ) |>
  rename(`Years` = years,
         `Staff_number` = No_ft_staff,
         `Rangking` = ranking,
         `Graduate number` = No_Graduate_SS,
         `Undergrad number` = No_Undergra,
         `Investment return(m, RMB)` = Investment_return,
         `Total invesment(m, RMB)` = T_investment,
         `Total contributions(m, RMB)` = T_contributions,) |>
  skim() |>
  yank("numeric") |>
  select(-hist, -complete_rate)

# Descriptive statistics for percentage variables
data |>
  select(ROI_invest_return_T_invest_PC, PE_T_assets_PCT, OE_T_expenses_PCT) |>
  rename(`ROI` = ROI_invest_return_T_invest_PC,
         `Porject expense` = PE_T_assets_PCT,
         `Operation expense` = OE_T_expenses_PCT) |>
  skim() |>
  yank("numeric") |>
  select(-hist, -complete_rate)

# Descriptive statistics for core variables
options(scipen = 999)
data |>
  select(years, No_ft_staff, ranking,
         No_Graduate_SS, No_Undergra,
         T_contributions) |>
  rename(`Years` = years,
         `Staff number` = No_ft_staff,
         `Rangking` = ranking,

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    `Graduate number` = No_Graduate_SS,
    `Undergrad number` = No_Undergra,
    `Total contributions(m, RMB)`= T_contributions,) |>
skim() |>
yank("numeric") |>
select(-hist, -complete_rate)

# Initial correlation analysis
cor(data$No_Graduate_SS,data$T_contributions)

data |>
  select(-Name_Eng) |>
  cor()

data |>
  select(-Name_Eng) |>
  cor() |>
  data.frame()

data |>
  select(where(is.numeric)) |>
  cor() |>
  data.frame() |>
  select(T_contributions) |>
  filter(abs(T_contributions) > 0.6)

data |>
  select(- Name_Eng) |>
  cor() |>
  data.frame() |>
  rownames_to_column("variables_1") |>
  pivot_longer(!variables_1, names_to = "variables_2",
               values_to = "correlation") |>
  filter(abs(correlation) > 0.6)

# Detailed correlation analysis for key variables
correlation <- data |>
  select(ranking, No_Graduate_SS, No_Undergra,
         No_ft_staff, years, T_contributions) |>
  cor() |>
  data.frame() |>
  rownames_to_column("variables_1") |>
  pivot_longer(!variables_1, names_to = "variables_2",
               values_to = "correlation") |>
  filter(abs(correlation) > 0.5) |>
  filter(variables_1 != variables_2) |>
  mutate(new_variables_1 = pmin(variables_1, variables_2),
         new_variables_2 = pmax(variables_1, variables_2) ) |>
  select(new_variables_1,new_variables_2, correlation) |>
  distinct() |>
  mutate(new_variables_1 = case_when(
    new_variables_1 == "No_Graduate_SS"~"Graduate number",
    new_variables_1 == "T_contributions"~"Contribution",

```

```

new_variables_1 == "No_ft_staff"~"Staff number",
new_variables_1 == "ranking"~"Ranking",
TRUE~new_variables_1
)) |>
mutate(new_variables_2 = case_when(
  new_variables_2 == "ranking"~"Ranking",
  new_variables_2 == "T_contributions"~"Contribution",
  new_variables_2 == "years"~"Years",
  TRUE~new_variables_2
)) |>
arrange(desc(abs(correlation)))

test1 <- cor.test(data$ranking, data$No_Graduate_SS, method = "spearman")
test2 <- cor.test(data$T_contributions, data$No_ft_staff, method = "spearman")
test3 <- cor.test(data$T_contributions, data$No_Graduate_SS, method = "spearman")
test4 <- cor.test(data$T_contributions, data$years, method = "spearman")
test5 <- cor.test(data$T_contributions, data$ranking, method = "spearman")

correlation$p_value <- c(test1$p.value, test2$p.value, test3$p.value,
                          test4$p.value, test5$p.value)

correlation |>
  kable(digits = 6)

# Visualizing correlations with ggplot
data |>
  ggplot(aes(x=log(T_contributions), y=ranking)) +
  geom_point() +
  geom_smooth(method = "lm") +
  labs(x = "Contribution(log)", y = "Ranking", title = "Correlation")

data |>
  ggplot(aes(x=log(T_contributions), y=No_ft_staff)) +
  geom_point() +
  geom_smooth(method = "lm")+
  labs(x = "Contribution(log)", y = "Staff number", title = "Correlation")

data |>
  ggplot(aes(x=log(T_contributions), y=No_Graduate_SS)) +
  geom_point() +
  geom_smooth(method = "lm")+
  labs(x = "Contribution(log)", y = "Graduate number", title = "Correlation")

data |>
  ggplot(aes(x=log(T_contributions), y=years)) +
  geom_point() +
  geom_smooth(method = "lm")+
  labs(x = "Contribution(log)", y = "Foundation age", title = "Correlation")

data |>
  ggplot(aes(x=log(T_contributions), y=No_Undergra)) +
  geom_point() +
  geom_smooth(method = "lm")+
  labs(x = "Contribution(log)", y = "Undergrad number", title = "Correlation")

```



```

# Descriptive statistics for Mulan financial variables
options(scipen = 999)
Mulan |>
  select(`Year-updated names`, `No. of full time staff`,
        `Investment return`, `Total investments`,
        `Total revenues`, Contributions,) |>
  rename(`Year range` = `Year-updated names`,
        `Staff number` = `No. of full time staff`,
        `Investment return(m, RMB)` = `Investment return`,
        `Total investments(m, RMB)` = `Total investments`,
        `Total revenues(m, RMB)` = `Total revenues`,
        `Contributions(m, RMB)` = Contributions) |>
  skim() |>
  yank("numeric") |>
  select(-hist, -complete_rate)

# Descriptive statistics for Mulan ROI
options(scipen = 999)
Mulan |>
  select(`ROI investment return/Total investments %`) |>
  rename(`ROI` = `ROI investment return/Total investments %`) |>
  skim() |>
  yank("numeric") |>
  select(-hist)

# Descriptive statistics for GPU financial variables
options(scipen = 999)
GPU |>
  select(Year, `No. staff`, Contributions, `Total revenues`,
        `Investment return`, `Total investments`) |>
  rename(`Staff number` = `No. staff`,
        `Contributions(m, USD)` = Contributions,
        `Total revenues(m, USD)` = `Total revenues`,
        `Investment return(m, USD)` = `Investment return`,
        `Total investments(m, USD)` = `Total investments`
  ) |>
  skim() |>
  yank("numeric") |>
  select(-hist, -complete_rate)

# Descriptive statistics for GPU ROI
options(scipen = 999)
GPU |>
  select(`ROI investment return/Total investments %`) |>
  rename(`ROI` = `ROI investment return/Total investments %`) |>
  skim() |>
  yank("numeric") |>
  select(-hist, -complete_rate)

# ggplot ROI histogram for Chinese universities
data |>
  ggplot(aes(x=ROI_invest_return_T_invest_PC)) +
  geom_histogram(bins = 5, color = "white", fill = "#0072B2") +

```

```

stat_bin(bins = 5, geom = "text", color = "black",
         aes(label = after_stat(count)), vjust = -0.5) +
labs(x = "ROI", y = "Count", title = "ROI distribution among the top 39 Chinese universities")

# ggplot histograms of undergraduate and graduate numbers
t.test(x=data$No_Graduate_SS,y=data$No_Undergra,paired = TRUE)
data |>
  ggplot(aes(x=No_Undergra)) +
  geom_histogram(bins = 15, color = "white") +
  stat_bin(bins = 15, geom = "text", color = "blue",
           aes(label = after_stat(count)), vjust = -0.5)

data |>
  ggplot(aes(x=No_Graduate_SS)) +
  geom_histogram(bins = 15)

# ggplot ROI time series for Mulan and GPU (10-year data)
Mulan |>
  ggplot(aes(x=`Year-updated names`,
            y=`ROI investment return/Total investments %`)) +
  geom_line() +
  labs(x = "Year", y = "ROI", title = "Analysis of ROI of Mulan for 10 years") +
  scale_x_continuous(labels = seq(2010, 2019, by = 1),
                    breaks = seq(2010, 2019, by = 1) ) +
  scale_y_continuous(labels = scales::percent)

GPU |>
  ggplot(aes(x=`Year`, y=`ROI investment return/Total investments %`)) +
  geom_line() +
  labs(x = "Year", y = "ROI", title = "Analysis of ROI of GPU for 10 years") +
  scale_x_continuous(labels = seq(2010, 2019, by = 1),
                    breaks = seq(2010, 2019, by = 1) ) +
  scale_y_continuous(labels = scales::percent)

Mulan_ROI <- Mulan |>
  mutate(Year = `Year-updated names`, university = "Mulan") |>
  select(Year, `ROI investment return/Total investments %`, university)

GPU_ROI <- GPU |>
  mutate(university = "GPU") |>
  select(Year, `ROI investment return/Total investments %`, university)

ROI <- rbind(Mulan_ROI, GPU_ROI)

# ggplot ROI comparison between Mulan and GPU (2010-2019)
ROI |>
  ggplot(aes(x=Year, y=`ROI investment return/Total investments %`,
            colour = university)) +
  geom_line() +
  labs(x = "Year", y = "ROI",
       title = "ROI time series for Mulan and GPU (2010-2019)") +
  scale_x_continuous(labels = seq(2010, 2019, by = 1),
                    breaks = seq(2010, 2019, by = 1) ) +

```

```
scale_y_continuous(labels = scales::percent) +  
geom_point()
```